disk_tripwire.sh — Companion Guide

Revision: 1 **Last modified:** 2026-09-24T11:59:07Z

Source: `constitution/scripts/codegraph/disk_tripwire.sh` (105 lines). Every statement below cites `disk_tripwire.sh:<line>`.

Overview

A bulk CodeGraph index grows its SQLite database without bound. On a nearly full volume that exhausts the disk (header, lines 5-9). `disk_tripwire.sh` is a watcher that samples the free space of the volume holding the database and, the moment free space drops below `--min-gib`, sends **SIGSTOP** to the indexer process — a reversible pause (resume with `kill -CONT <pid>`), never SIGKILL (lines 7-9, 97-98).

It is started by `codegraph_safe.sh` (header cross-refs, line 30); it can also be run by hand against an indexer that was started some other way.

Prerequisites

- `bash`, `df`, `python3` (line 29).
- `codegraph_safe_helper.py` in the same directory — used for the process identity check (`HELPER="$HERE/codegraph_safe_helper.py"`, line 35; `identity()`, line 67).
- A live indexer PID whose real `/proc` argv is a `node … /lib/dist/bin/codegraph.js` process (lines 12-14). A shell whose command line merely mentions `codegraph.js` is refused (line 14).

Usage examples

```
# Pause indexer PID 12345 if the DB volume drops below 20 GiB free,
# sampling every 30 s (default).
disk_tripwire.sh --pid 12345 --dir <project_root>/.codegraph \
                 --min-gib 20 --log /path/to/tripwire.log

# Faster sampling
disk_tripwire.sh --pid 12345 --dir <project_root>/.codegraph \
                 --min-gib 50 --interval 5 --log /path/to/tripwire.log

# After a trip: free space, then resume the paused indexer
kill -CONT 12345
```

Arguments (lines –)

Flag	Required	Meaning
`--pid <n>`	yes	Indexer PID. Must be an integer > 1 (lines 51-52) and pass the identity check (lines 69-74).
`--dir <path>`	yes	Any path on the DB volume; used as the `df` target. Must exist as a directory (line 59).
`--min-gib <n>`	yes	Threshold in GiB, integer, bounded to `[5,500]` (lines 53-57).
`--interval <s>`	no	Seconds between samples, positive integer, default `30` (lines 36, 58).
`--log <file>`	yes	Append-only log (line 48). On trip, a marker file `<log>.TRIPPED` is created (line 99).

Any other argument: `disk_tripwire: unknown argument` and exit 2 (line 44).

Test-only environment variable

`CG_SAFE_TEST_FREE_BYTES` — can only **lower** the measured free bytes: `avail = min(measured, override)` (lines 19-20, 85-86). It cannot fake extra space. A non-numeric value is ignored (line 86).

Exit codes (header lines – , verified against code)

Code	Meaning	Source
0	Indexer ended (PID gone, zombie `Z`, or dead `X`) with no trip	lines 79-82
2	Usage error or refused to arm: missing `--log`, bad/≤1 PID, non-integer or out-of-range threshold, bad interval, missing dir, unknown argument, target is not a codegraph indexer	lines 44, 47, 51-59, 70-73
3	PID changed identity while watched (PID reuse) — no signal sent; **also** returned when `kill -STOP` itself fails	lines 90-94, 102-103
10	TRIPPED: free < threshold, indexer SIGSTOPped, marker written	lines 96-100

Note: the header documents exit 3 only as "pid changed identity"; the code also uses 3 when the STOP signal fails (line 103).

Edge cases

- **Missing `--log`** is checked first and writes to stderr only (line 47); every later refusal is also appended to the log via `say` (lines 48-59).
- **`--interval 0`** is refused (line 58).
- **`df` output unreadable** (empty or non-numeric): logs `WARN df unreadable`, sleeps, retries — does not trip and does not exit (line 84).

- **PID reuse:** the identity string captured at arm time (`ID0` , line 69) is re-read immediately before signalling; if it differs the script refuses to signal and exits 3 (lines 88-93). Identity is only re-checked on the trip path, not on every sample (lines 87-89).
- **Process state** is read from field 3 of `/proc/<pid>/stat` after the last `)` so command names with spaces or parentheses do not break parsing (lines 61-66).
- A stopped (`T`) indexer is not treated as gone; only `-` , `Z` , `X` end the watch (line 79).

Internal behaviour

1. Parse flags (lines 38-46); validate `--log` , PID, threshold, interval, dir (lines 47-59).
2. Capture identity via `python3 codegraph_safe_helper.py identity <pid>` ; the output must begin with `OK` (lines 69-74). Log `ARMED …` (line 75).
3. Loop (lines 78-105): read process state; exit 0 if gone. Read `df -B1 --output=avail <dir>` (line 83); apply the test override; log `avail_bytes=… min_bytes=…` every sample (line 87).
4. If `avail < min_gib × 1073741824` (lines 77, 88): re-check identity, send `kill -STOP` , log `TRIPPED …` , touch `<log>.TRIPPED` , exit 10.
5. Otherwise `sleep <interval>` and repeat.

Log line format: `HH:MM:SS <message>` (`date +%T` , line 48).

Related scripts

Guide	Relation
codegraph_safe	Starts this watcher for index/sync runs
codegraph_safe_helper	Provides the `identity` subcommand used here
codegraph_guard	Refuses a second writer against a live lock holder
index_watch	Progress-proven watchdog for long runs
codegraph_runner_patches	Patched runner used for bulk runs
fk_index_patch	Runner patcher
fk_cascade_probe	Bulk-window hazard probe
codegraph_scope_guard	Scope proof before "ready"
scope_render	Generates the effective scope
codegraph_mcp_probe	MCP reachability probe
lumen_verify	Semantic-index health check
gen_lumenignore	Semantic-index scope generator
codegraph_lumen_overview	Overview of the indexing tooling

Tests: the header cites `tests/test_tripwire.sh` (line 30); the file exists in `scripts/codegraph/tests/`. UNCONFIRMED: its content and coverage were not read for this guide.

Last verified

2026-09-24 — read `disk_tripwire.sh` lines 1-105 in full; `bash -n disk_tripwire.sh` passed. The script was NOT executed (a live indexer holds the project DB).